	Marwadi University **Faculty of Engineering and Technology** **Department of Information and Communication Technology**	
Subject: DSIP (01CT1513)	**AIM: Image Smoothing and Sharpening**	
Experiment No: 09	**Date:**	**Enrolment No: 92301733041**

 # write a python program to simulate smoothening and sharpening operation on image using spatial filter import cv2 import numpy as np # Load the image image = cv2.imread(“image_smooth.jpg")

Define the Gaussian kernel for smoothing

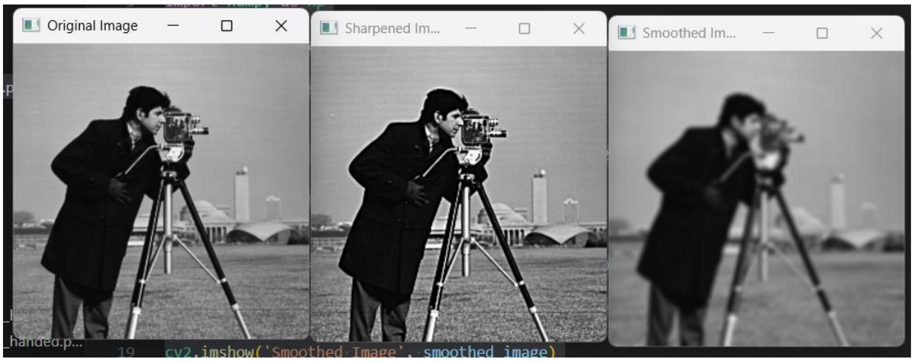
# Smoothing smoothed_image =
cv2.GaussianBlur(image, (7, 7), 2)

Define a sharpening kernel

Sharpening sharpened_image = cv2.addWeighted(image, 1.7,
smoothed_image, -0.7, 0)

Display the original image, smoothed, and sharpened images
cv2.imshow('Original Image', image) cv2.imshow('Smoothed
Image', smoothed_image) cv2.imshow('Sharpened Image',
sharpened_image) cv2.waitKey(0) cv2.destroyAllWindows()

Output:



write a python program to simulate smoothening using Averaging linear filters

import cv2 import numpy as np # Load the image image = cv2.imread("
image_smooth.jpg")

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# Define the size of the Averaging filter kernel

# Apply Gaussian Blur for better smoothing and clarity smoothed_image
= cv2.GaussianBlur(image, (7, 7), 1.5)

# Display the original and smoothed images
cv2.imshow('Original Image', image)
cv2.imshow('Smoothed Image', smoothed_image)
cv2.waitKey(0) cv2.destroyAllWindows()

output:
```